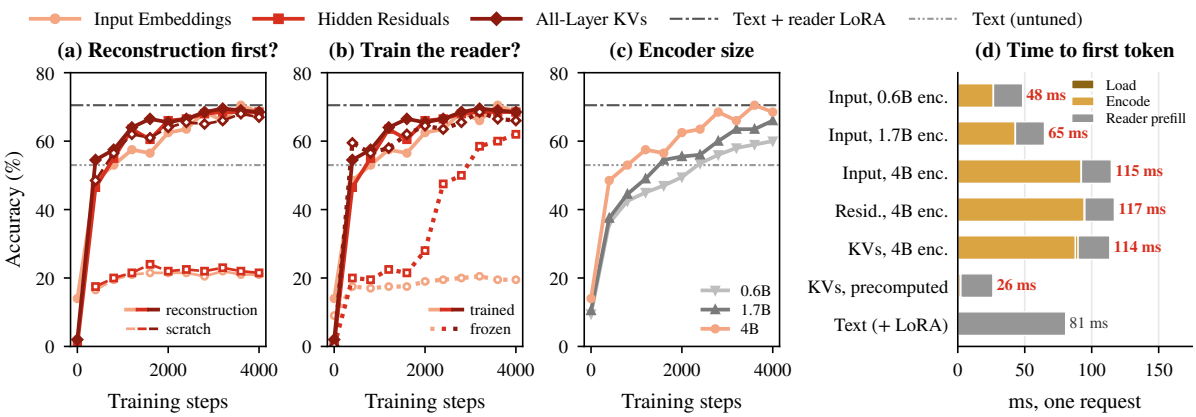
Input Embeddings
Hidden Residuals
All-Layer KVs
Text + reader LoRA
Text (untuned)
(a) Reconstruction first?
(b) Train the reader?
(c) Encoder size
(d) Time to first token
Accuracy (%)
Training steps
reconstruction
scratch
trained
frozen
0.6B
1.7B
4B
Load
Encode
Reader prefill
Input, 0.6B enc.
Input, 1.7B enc.
Input, 4B enc.
Resid., 4B enc.
KVs, 4B enc.
KVs, precomputed
Text (+ LoRA)
48 ms
65 ms
115 ms
117 ms
114 ms
26 ms
81 ms
ms, one request